

Gemini Health — Context Report

Purpose

Gemini Health is a lightweight performance dashboard designed to import activity data (CSV bulk or single GPX), store activities in a file-based H2 database, compute training metrics (TRIMP, CTL/ATL/TSB), and visualize activities and training load in a web UI.

Primary Endpoints

- GET /api/athletes — list athlete profiles
- POST /api/athletes — create athlete (body: { name, primarySport, city, state, country })
- GET /api/athletes/{id}/activities — list activities for athlete
- POST /api/athletes/{id}/upload/csv — upload bulk CSV activities
- POST /api/athletes/{id}/upload/gpx — upload single GPX workout
- GET /api/athletes/{id}/performance — computed CTL/ATL/TSB timeline
- Diagnostic: GET /api/diagnostic/stats and GET /api/diagnostic/athlete/{id}/activities/count

Data Model Summary

- AthleteProfile: id, name, stravaId, avatarUrl, city/state/country, primarySport, weeklyDistanceGoal, restingHr, maxHr, ftp, activities
- Activity: id, stravaActivityId, athlete (many-to-one), name, type, startDate, distance (km), movingTime, elapsedTime, totalElevationGain, averageHr, maxHr, averageSpeed, averageWatts, trimp, streamJson

Important Behaviour Notes

- CSV import uses fuzzy header mapping and supports multiple date formats; rows may be skipped if malformed/duplicate/unparseable.
- GPX import parses trackpoints, builds a stream JSON, aggregates distance and elevation and computes average/peak heart-rate.
- Performance calculations: TRIMP is computed per activity; CTL/ATL are EWMA over 42 and 7 days respectively.

How an AI agent can use this context

- Use the endpoint list to query the system for available athletes and their activities.
- After an import, verify database persistence via diagnostic endpoints and request /api/athletes/{id}/performance for training insights.
- Use streamJson (if available) for fine-grained HR zone calculations.

Operational Notes

- The app uses H2 file database located under `./data (db.mv.db)`.
- Uploaded GPX creates an Activity and returns its payload; the frontend uses that to show a success banner and activity details.

Developer Notes — Upload UX change

- The frontend now prompts for athlete name when you attempt to upload a file and no athlete is selected. It will create the athlete if it does not exist and then proceed with the upload.
- To create an athlete manually via API: `POST /api/athletes` with JSON `{ "name": "Athlete Name" }`.

Quick commands

Start the application:

```
java -jar target/dashboard-0.0.1-SNAPSHOT.jar
```

Create an athlete (example with PowerShell):

```
Invoke-RestMethod -Uri 'http://localhost:8080/api/athletes' -Method Post  
-ContentType 'application/json' -Body '{"name":"Jane Athlete"}'
```

Check diagnostic status:

```
curl http://localhost:8080/api/diagnostic/stats
```

Converting to PDF

If you want a PDF version of this report you can:

- Open `context_report.md` in an editor and export/print to PDF (VSCode, Typora, etc.), or
- Use `pandoc` to convert: `pandoc context_report.md -o context_report.pdf`, or
- Use the included Java ReportGenerator located at `src/main/java/com/geminihealth/dashboard/util/ReportGenerator.java` (requires Apache PDFBox dependency):
 - Build: `mvn clean package -DskipTests`
 - Run generator with classpath including PDFBox jars (example):

```
java -cp "target/classes;C:\Users\<you>\.m2\repository\org\apache\pdfbox\pdfbox\2.0.29\pdfbox-2.0.29.jar;C:\Users\<you>\.m2\repository\org\apache\pdfbox\fontbox\2.0.29\fontbox-2.0.29.jar;C:\Users\<you>\.m2\repository\commons-logging\commons-logging\1.2\commons-logging-1.2.jar" com.geminihealth.dashboard.util.ReportGenerator
```

(If the Java generator emits nested text errors, prefer the `pandoc`/editor approach.)

If you want, I can:

- Wire a small UI modal (instead of prompt) to collect more athlete details on upload (city, sport, resting HR), and
- Automatically generate a PDF server-side on upload and return it in the response (requires adding a PDF endpoint and ensuring safe PDF generation).

Tell me which of these you'd like next and I will implement it.